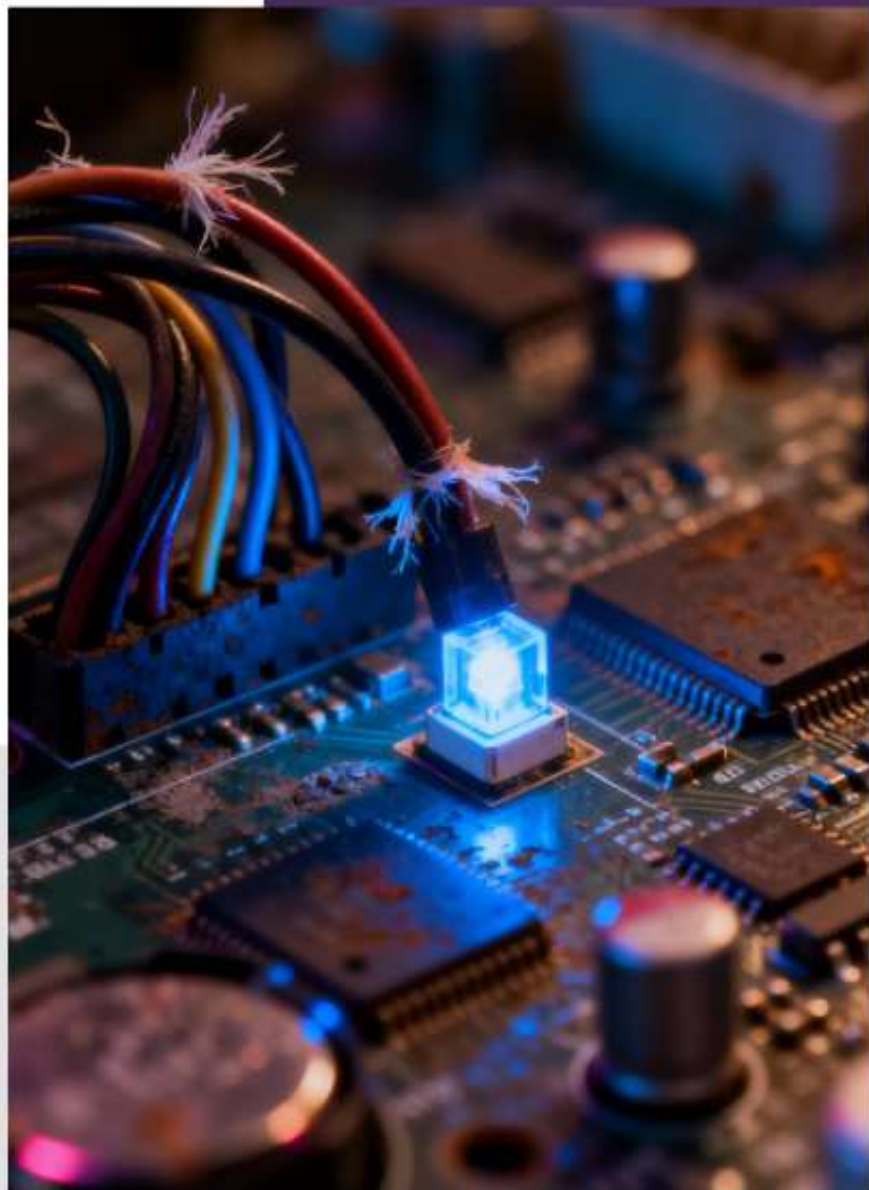




PathShala AI: Offline AI Architecture

Democratizing education in rural schools through high-performance edge computing and 100% offline AI assistance

Bridging the digital divide with quantized Llama-3.2 models and WebGPU acceleration.

Theme: A Teacher's Narration of the Problem



Real Problems Come From The Classroom

- Most Solutions Designed Top-Down Miss Daily Realities
- This Theme Centers The Teacher's Lived Experience
- Classroom Issues Emerge Continuously During Lessons



Highlights The Implementation Gap

- Training Shows What To Do But Not How To Do It Live
- Teachers Struggle To Apply Methods In Chaotic Settings
- Theme Exposes Mismatch Between Training And Practice

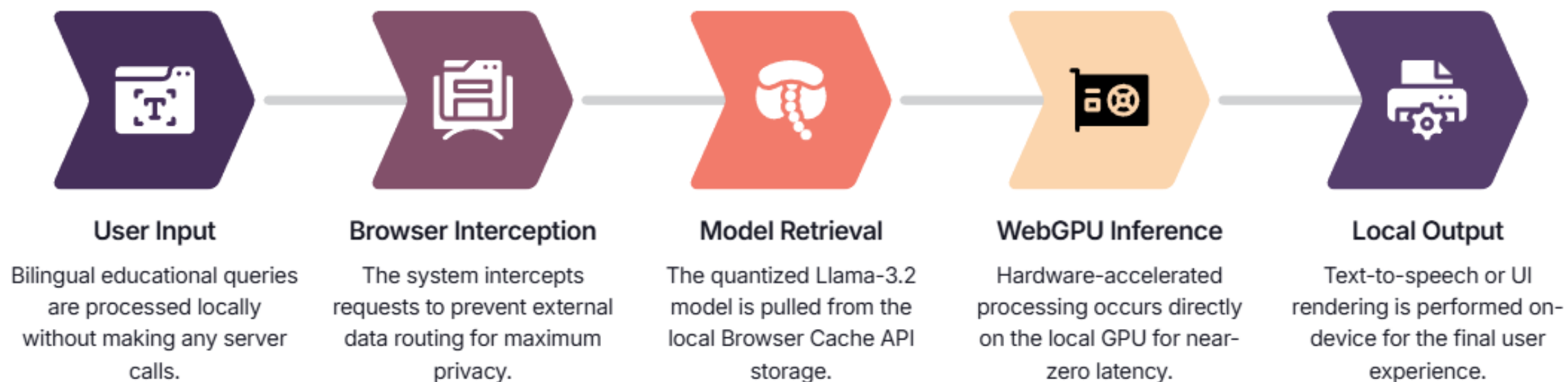


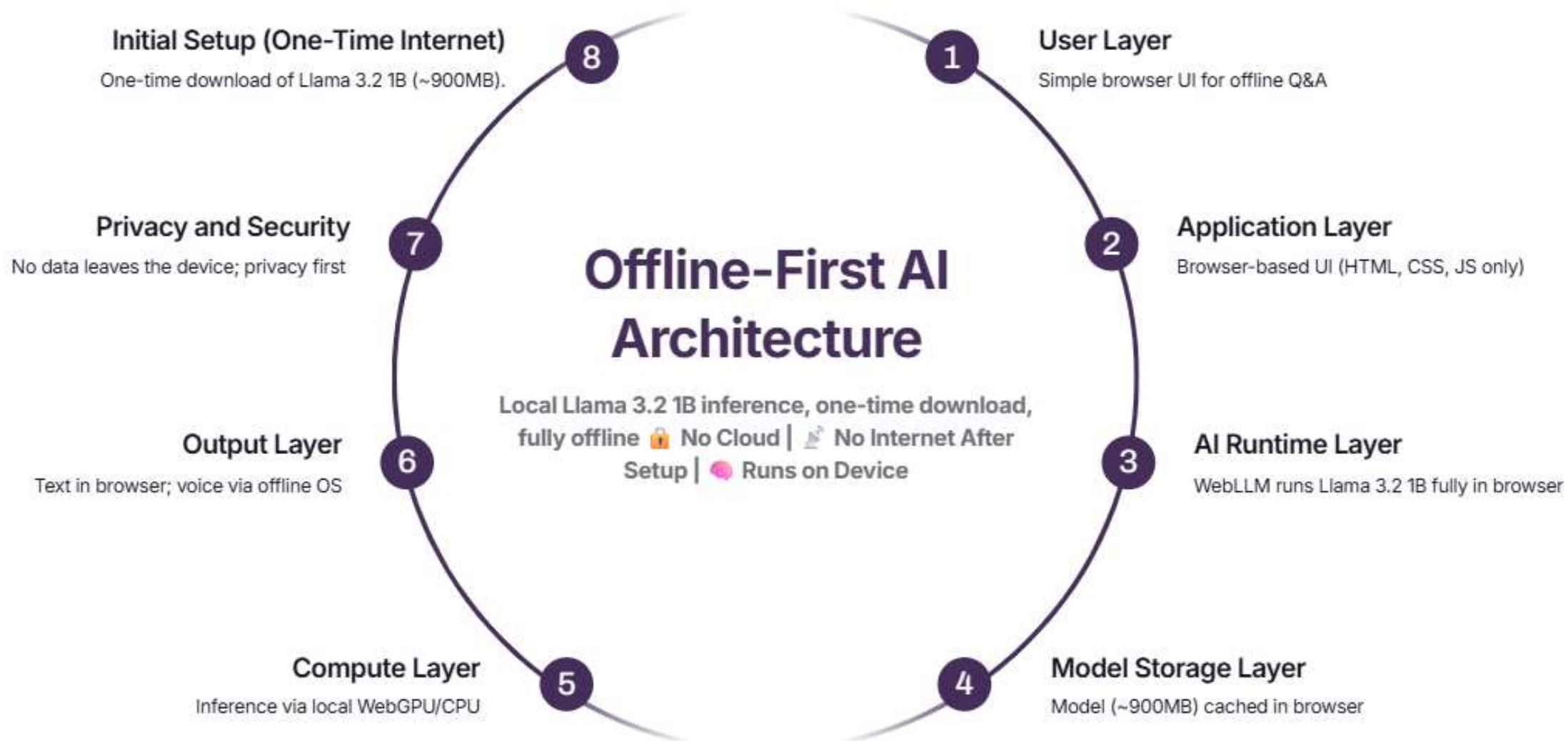
Focus On Just-In-Time Support

- Help Often Arrives Weeks Later Via Inspections Or Visits
- Delayed Support Misses The Moment The Problem Occurs
- Demand Immediate, Offline Tools That Assist In The Moment

Offline-First Data Flow

A sophisticated zero-latency pipeline that executes high-performance inference entirely within the local browser environment





One-Time Download for Forever Offline Use

Download once; runs offline for
low-connectivity schools

Model downloaded once (~900MB)

Single download size: approximately 900MB



Stored securely in browser cache

Local storage in the browser cache for safety



Works without Internet

Full AI functionality offline after download



Ideal for rural and low-connectivity schools

Designed for environments with limited internet



The Technical Stack

Leveraging cutting-edge web technologies to turn standard hardware into high-performance AI workstations



WebLLM (Apache TVM) serves as the core **Inference Engine** for local model execution.



WebGPU API enables direct **Hardware Access** for accelerated graphics and compute processing.



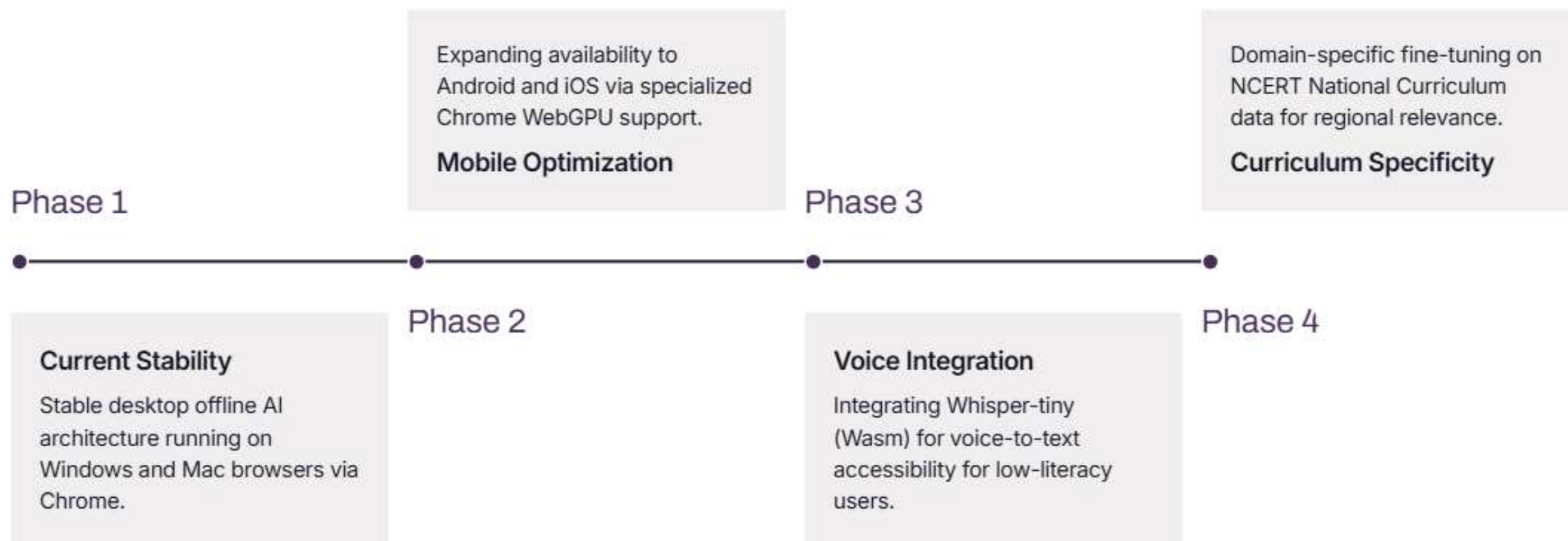
Vanilla HTML5/CSS3 provides a **Zero Bloat** user interface for maximum performance on old hardware.



Llama-3.2-1B-Instruct model provides high-quality bilingual educational assistance.

Future Strategic Roadmap

The evolution of PathShala AI from desktop stability to mobile accessibility and curriculum-specific fine-tuning





Democratized AI for All

By combining edge computing with local language models, PathShala AI is building a more equitable educational landscape where intelligence is no longer restricted by internet access.

Ensuring quality education reaches every corner of the world, regardless of connectivity.